

PHYSICS OF SEYFERT GALAXIES

IASc. SRFP-2014 Project Report

by

SWAYAMTRUPTA PANDA

IASc. SRFP Regd. No.: PHYS550-I

National Institute of Technology, Rourkela

Under the Guidance of

PROF. PRAJVAL SHASTRI

Indian Institute of Astrophysics, Bangalore

ACKNOWLEDGEMENT

First of all, I express my heartiest gratitude to the Indian Academy of Sciences for finding me worthy of doing this summer project under Prof. Prajval Shastri at IIA Bangalore. I'm also grateful to the people at IIA Bangalore for their hospitality and services. My sincere thanks to Prof. Shastri who is seconded to none in what she does and the way she put the idea into my head that got me driven into this work. She has been there for me the entire time, be it some nitty-gritty issue or putting forward newer details into the topic or the field of astronomy and astrophysics itself. Her in-depth knowledge about the field and stimulating discussions providing insights to the matter proved to be vital in the successful completion of the work and bringing it to its present shape. I'm grateful to Ms. Maitrayee Gupta who has had a pivotal role in this work. Be it her assistance, her compassion and knowledge about the work, it has been instrumental in helping me in the entire period of the project. I also would like to thank Mr. Vikram Radhakrishnan and Mr. Sundar M N for their support and involvement in the work.

I'd also like to acknowledge the following:

1. **NED:** This research has made use of the NASA/IPAC Extragalactic Database (NED) which is operated by the Jet Propulsion Laboratory, California Institute of Technology, under contract with the National Aeronautics and Space Administration.)
2. **HEASARC:** This research has made use of data, software and/or web tools obtained from NASA's High Energy Astrophysics Science Archive Research Center (HEASARC), a service of Goddard Space Flight Center and the Smithsonian Astrophysical Observatory.
3. **ADS:** This research has made use of NASA's Astrophysics Data System Bibliographic Services.
4. **ZORROASTER:** This research has made use of Hartmut Winkler's Catalogue of Seyfert galaxies based on the SDSS

spectra (<http://www.uj.ac.za/EN/Faculties/science/departments/physics/Pages/ZORROASTER.aspx>)
5. **R-Project:** The plots and analysis for this research used software tools from the R Project. (<http://www.r-project.org/>)
6. **TopCat:** Compilation of the AGN data used in this research made use of the software tools of TopCat. Astronomical Data Analysis Software And Systems XIV ASP Conference Series, Vol. 347, 2005 P. L. Shopbell, M. C. Britton, and R. Ebert, eds. 2005ASPC...347...29T
7. This research made use of Ned Wright's Cosmological calculator

(<http://www.astro.ucla.edu/~wright/CosmoCalc.html>)

Lastly, I express my deep gratitude to the Almighty and my parents for their blessings and support when I needed them the most.

Date: 17 Jul 2014

Swayamtrupta Panda

IASc. SRPF Regd. No.: PHYS550-I

National Institute of Technology, Rourkela

ABSTRACT

Active Galactic Nuclei (AGN) are accreting supermassive black holes. AGNs exhibit prodigious luminosities from their nuclei. They are classified as “Radio Loud” and “Radio Quiet” based on the relative amount of radio to optical emission. Seyfert galaxies are radio-quiet AGNs which are typically found in spiral galaxies and have low-powered jets. In Seyferts unlike the higher luminosity Quasars, the host galaxies are clearly visible, because the Seyferts are relatively nearby. This study focuses on a sample of 128 Active Galactic Nuclei for which a survey to determine the emission-line morphology is on-going. We characterize the physical properties of the sample with multi-wavelength data.

CONTENTS

1. Introduction to Active Galactic Nuclei	5
2. AGN Classification	7
3. S7 Survey Sample	9
4. Systematics	12
5. Conclusions	19
6. References	20

INTRODUCTION TO ACTIVE GALACTIC NUCLEI

Active Galactic Nuclei are luminous and compact cores of galaxies which are fuelled by accreting supermassive black holes. The active phase is a typical feature of active galaxies that is not observable in other normal galaxies. (Blandford, 1990)

“Because their high luminosities and distinctive spectra make them relatively easy to pick out, they are disproportionately represented in our tally of known high redshift sources. The fact that the luminosity of AGNs is such a strong function of redshift suggests that there is something special about youthful galaxies that promote the creation of active nuclei.” (Krolik, 1999)

Certain salient features of AGNs are:

AGNs power output ranges from about 1% to $\sim 10^4$ times the stellar luminosity of typical galaxy. There could be obscuration of the central accretion disk system by thick dust, resulting in far less measured luminosity than its true luminosity. Relativistic beaming (the process by which relativistic effects modify the apparent luminosity of the bipolar jets that are moving at speeds close to the speed of light) can substantially distort the angular distribution of light from the AGNs along with the selection effect (wherein there involves a selection based on the observation of those entities whose radiating materials moves towards us). (Urry & Padovani, 1995).

The spectral energy distribution of active galaxies covers the full-wavelength range (from radio waves to Gamma-rays) including both thermal and non-thermal (synchrotron and inverse-Compton scattering) radiation.

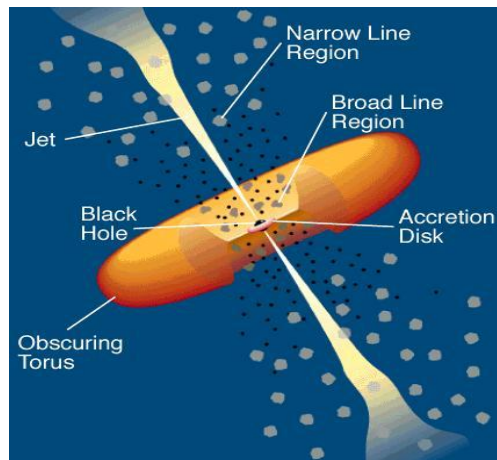


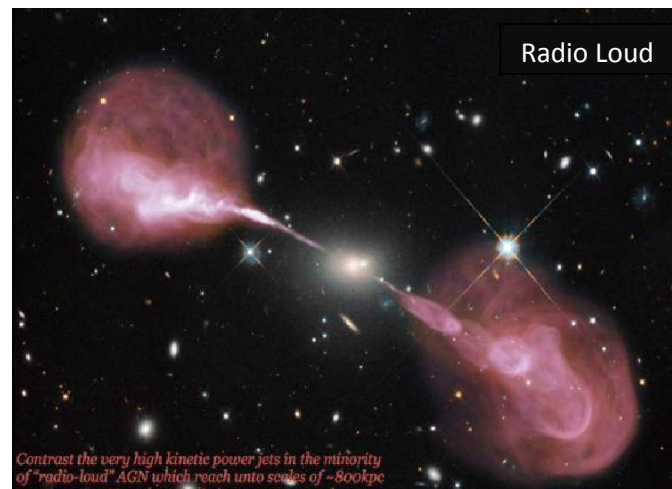
Fig.1. AGN Model (Urry & Padovani, 1995)

According to the standard AGN Model, there exists a super-massive black hole at the center of the AGN that is accreting matter from the galaxy (Urry & Padovani, 1995). This accretion leads to creation of the bipolar jets radiating synchrotron emission. There is evidence of an obscuring torus made up of gases and dust which causes extinction of the AGN continuum. The spectra of AGN contain broad permitted emission lines (FWHM ~ 1000 km/s) and narrow permitted and

forbidden lines (FWHM $\sim 300\text{--}800$ km/s). AGN emission lines are very prominent (equivalent widths $\sim 100\text{\AA}$) which makes the AGN spectra reflect a great contrast to the spectra of most stars and galaxies (where lines are generally relatively weak and predominantly in absorption). We almost always see Ly α , the balmer lines, the CIV 1549 doublet, [OIII] 5007, and several others that are generally weak. Most AGNs can be seen to vary on a variety of timescales. In some cases, fluctuations of a factor of two have been seen from night to night, and cumulative changes of factors of 100 have occurred over year timescales. Strong variability is also very strongly correlated with strong polarization, compact radio structure, and strong high-energy γ -ray emission.

AGN CLASSIFICATION

AGNs are broadly categorized as being “Radio Quiet” or “Radio Loud” based on the power of their radio emitting synchrotron jets. In the case of Radio Loud AGNs these jets are highly relativistic which extend to distances on the scale of a megaparsec that is, much more than the extent of the host galaxy itself. But in the case of Radio Quiet AGNs, these jets are limited inside the host galaxies. The ratio of the Radio flux density to that of the optical flux density is taken to be a measure of the AGN’s radio loudness (*Kellerman et. al. 1995*).



NGC1068

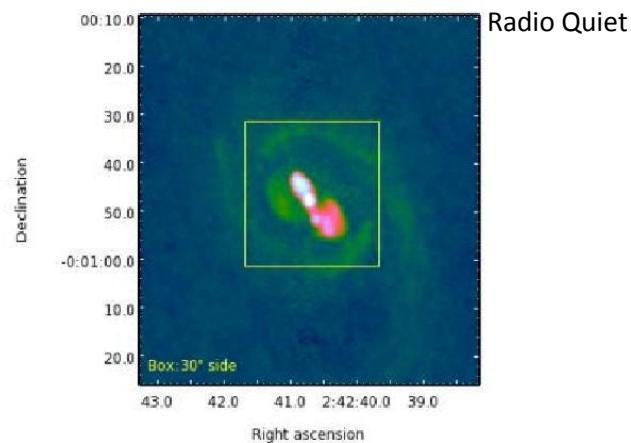


Fig.2 Radio Loud AGN (Hercules A) and Radio Quiet AGN (NGC 1068)

Seyfert galaxies (a sub-class of AGNs):

Seyfert galaxies are nearby low-luminosity ($M_V > -23$) AGNs typically in spiral host galaxies with strong emission lines in their optical spectra. The Seyferts are classified as Type 1 and Type 2.

The distinction between the Sy1 and Sy2 types lies in the nuclear **broad** components in the permitted lines (FWHM > 1000 km/s) (and narrow forbidden) emission lines in the spectra of Sy1, while in the Sy2 the nuclei tend to have equally wide lines for the permitted and forbidden lines without the very broad (± 5000 km/s) wings.



Fig.3. NGC 4151, a Seyfert Type 1 galaxy

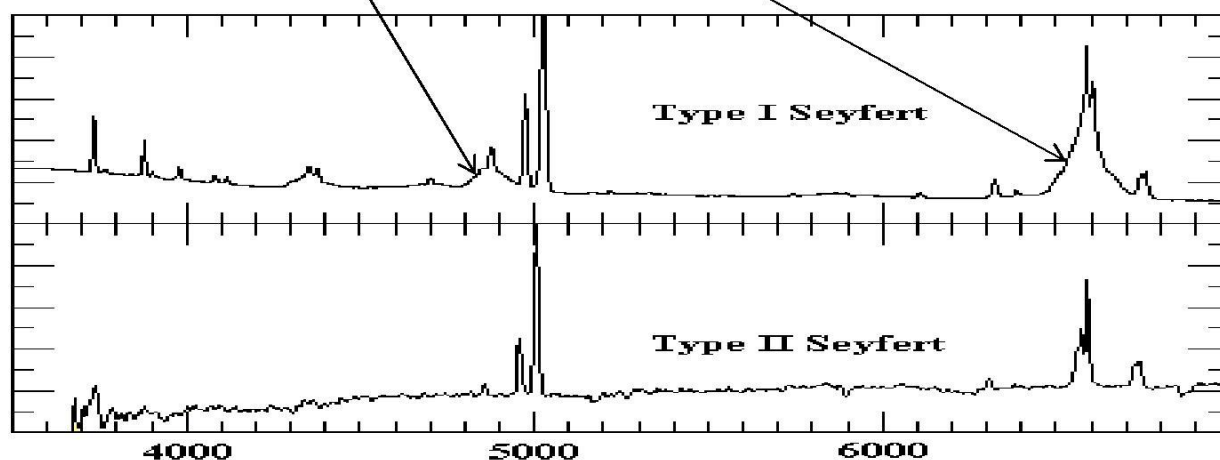


Fig.4 Type 1 and Type 2 Seyfert galaxies spectra

There is broad empirical support for the idea that Seyferts of type2 are those with their accretion disc system edge-on w.r.t. our line of sight, so that the toroidal gas and dust obscures the broad-line emitting clouds from our view (*Antonucci 1993*)



Fig.5. NGC 1068, a Seyfert Type 2 galaxy

S7 SURVEY SAMPLE

This project involved characterizing the physical properties of the sample of the S7 survey using multi-wavelength data.

- S7 - Siding Spring Southern Seyfert Spectroscopic Snapshot Survey
- The S7 survey is a survey of 128 of the most luminous Seyfert galaxies in the local universe, being conducted using the Wide Field Spectrograph (WiFeS) on the ANU 2.3m telescope.
- The targets are taken from Veron-Cetty and Veron Catalogue which is the most comprehensive list of AGNs



Fig.6. Wide Field Spectrograph (background) mounted on the ANU 2.3m telescope

WiFeS is mounted on the 2.3m ANU telescope at Siding Spring Observatories. It is an Integrated Field/Imaging Spectrograph that records spectrum for each pixel of the region that is imaged and produces a data cube. (*Dopita, M. et. al. 2010*)



Fig.7. ANU 2.3m telescope at Siding Spring

The S7 team selected these 128 AGNs on the basis of the following criteria:

- *Declination* < 10 degrees (for not too low elevation at 2.3 m Telescope, Siding Spring)
- *Galactic latitude* > |20| degrees (to avoid heavy Galactic extinction)
- *Redshift* < 0.02: so that have several beam elements across the host galaxy, & SII is in the spectral range.
- *Radio flux density at 20 cm* > 20 mJy (for radio follow-up observations)

Parameters taken from Veron-Cetty and Veron Catalogue:

- *Redshift*
- *Radio flux density at 6cm*
- *Radio flux density at 20cm*
- *BV Color Index*

Other data for the sample targets that were compiled:

- *IRAS flux density at 12 μ m, 25 μ m, 60 μ m and 100 μ m*

The IRAS flux densities are taken from IRAS (Infrared Astronomical Satellite) HEASARC (High Energy Astrophysics Science Archive Research Center) FSC (Faint Source Catalogue) and PSC (Point Source Catalogue) using similar criteria used to select the 128 AGNs from the Veron-Cetty & Veron Catalogue. The obtained list of IRAS objects are then matched with our S7 catalogue using the software tool **TopCat**. We were able to get flux densities at the 12 μ m, 25 μ m, 60 μ m and 100 μ m for 106 objects which are there in the S7 catalogue. The corresponding flux densities were added to the existing list of 128 AGNs.

- *Flux density (B-band)*

We first calculate the apparent B magnitude using the relation:

$$m_B - m_V = B - V$$

where B-V = BV color index;

(i)

m_V = app. magnitude in V band ($\sim$ Visual); and
 m_B = app. B magnitude

Then using the relation:

$$m_B - m_B^{\text{Vega}} = -2.5 \log(S_B/S_B^{\text{Vega}}) \quad (\text{ii})$$

we calculated the value for S_B i.e., the B flux density.

$m_B^{\text{Vega}} = B$ Magnitude for Vega = 0;

$S_B^{\text{Vega}} = B$ flux for Vega = 4130 mJy

- *Radio Loudness*

With the Radio Flux density at 6cm (corresponding to 5GHz) and the B flux density obtained using the above relations, we calculated the Radio Loudness which is given as

$$R_L = (S_{5\text{GHz}}/S_B) \quad (\text{iii})$$

We define an AGN as Radio Loud if the $R_L > 10$.

- *Radio Spectral Index*

For the objects that didn't have a flux at 6cm but instead had a value for the flux density at 20cm (which is equivalent to flux density at 1.4 GHz), we used the relation stated below to calculate the corresponding value for the flux density at 6cm:

$$S_{5\text{GHz}} = S_{1.4\text{GHz}} \times (5\text{GHz}/1.4\text{GHz})^{-0.7} \quad (\text{iv})$$

where $0.7 = \alpha$ = Radio Spectral Index [Using the relation: $S_\nu \sim \nu^{-\alpha}$]

- *Luminosity distance*

Assuming the Concordance Cosmology model parameters model with $\Omega_M = 0.286$, $\Omega_{\text{vac}} = 0.714$, Hubble parameter $H_0 = 69.6$, and using the values of redshift for the objects as obtained from the Veron-Cetty & Veron Catalogue, we are able to get the corresponding values for Luminosity distance using Ned Wright's cosmology calculator for each of 128 objects.

- *Luminosities from all the flux densities*

Using the values obtained for the Luminosity distance and the Flux densities at Radio, Infrared, B-band and V-Band already there in the S7 list, we are able to calculate the corresponding Radio, Infrared, B-band and V-band luminosities using the relation:

$$L = 4\pi r^2 F \quad (\text{v})$$

where L = luminosity ($\text{Mpc}^2 \text{ Jy}$)

r = luminosity distance (Mpc)

F = Flux density (Jy)

SYSTEMATICS

We have 128 objects with the parameters from Veron-Cetty & Veron Catalogue with the added parameters like the IRAS flux densities for 106 objects out of the 128. Also we have calculated certain parameters using the existing ones.

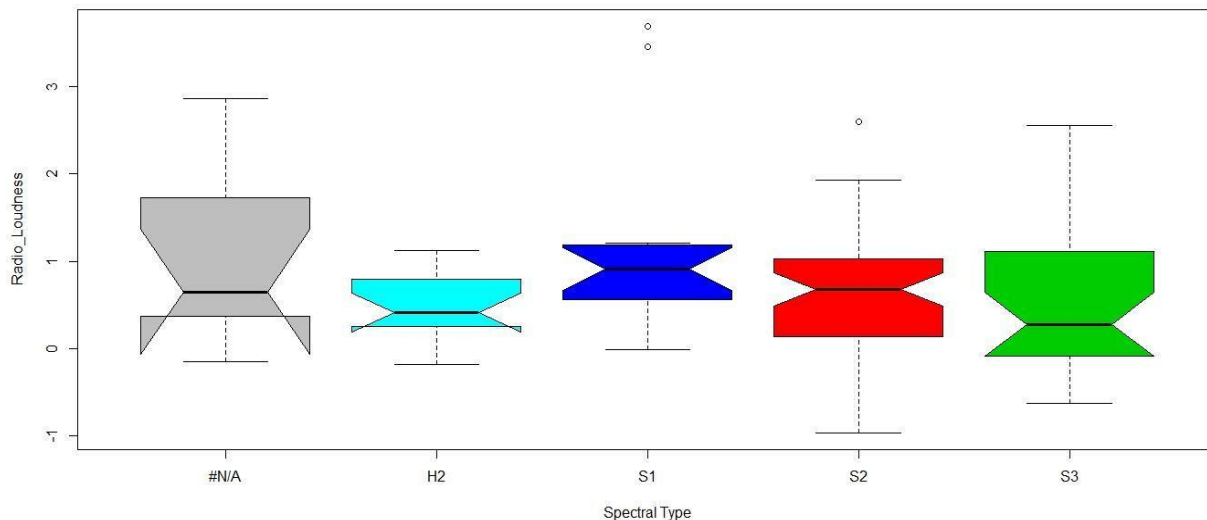
Using the Object_type provided in the S7 and the criteria stated in the HEASARC VeronCat, we have re-categorized the 128 objects as **S 1, S 2, S 3 (or LINERS) and H II** (this classification is based on Veron's classification). The remaining objects which didn't fall in any of these classes were put together as **N/A**.

We attempted to refine the classification based on the Spectral-Type by comparing the S7 list with **Hermut Winkler's Catalogue** of Seyfert galaxies "Zorroaster" (classification is based using SDSS spectra) and found 69 matches. We used the Spectral-Type according to Zorroaster for those 69 matches and for the rest used the VeronCat classification. Finally we pooled in all of the 128 to broadly classify them as H2, S1, S2, S3 and N/A.

Parameter distributions:

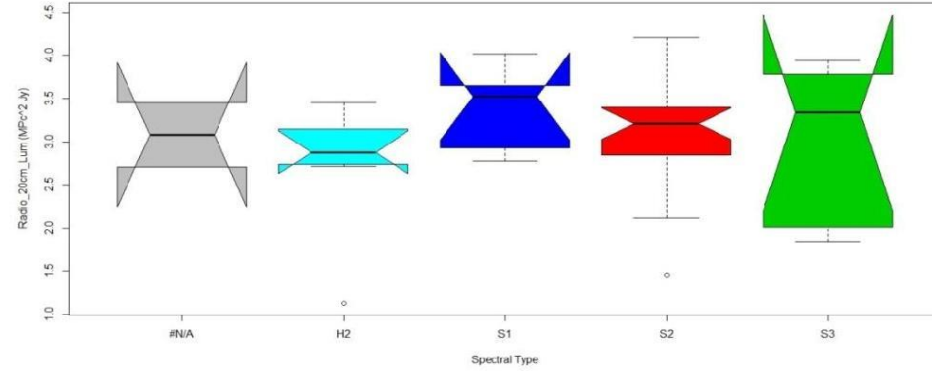
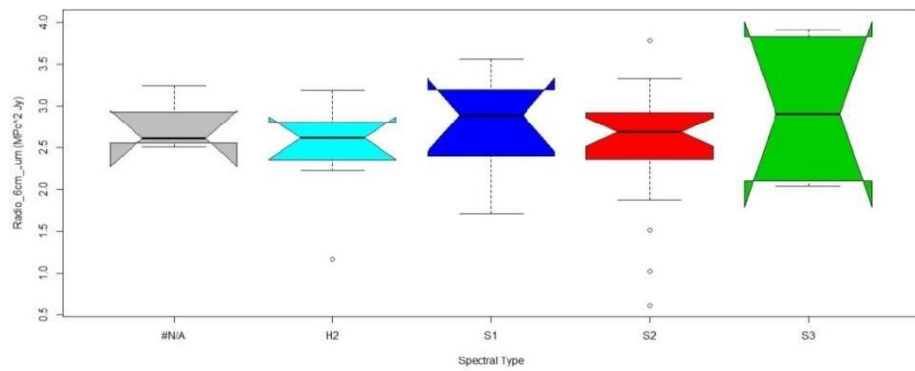
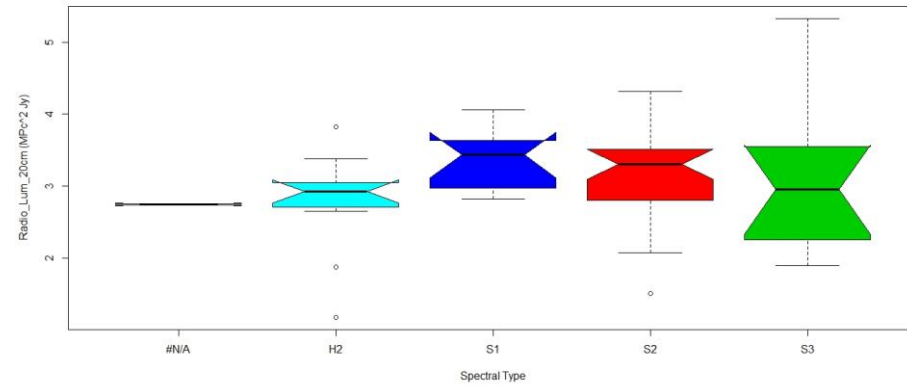
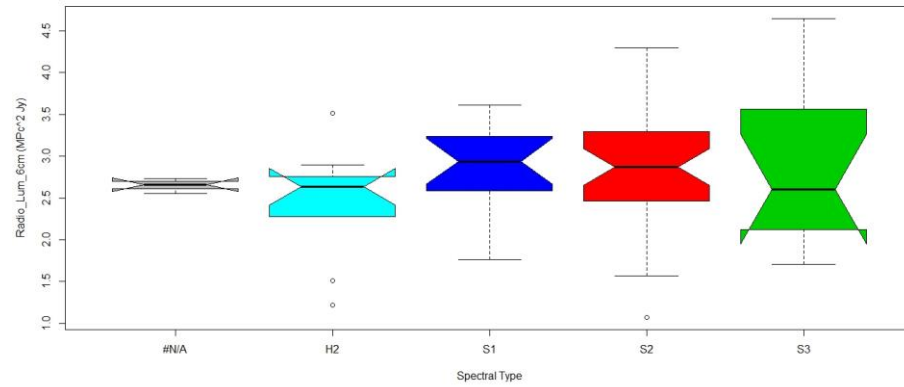
1. We have plotted the distribution of different parameters of the 128 objects in the S7 sample using Notched Box and Whisker plots (*Doyle, weblink*)

Radio Loudness for the different Spectral type:



This Box and Whisker plot shows the distribution of the objects' Radio Loudness for the different spectral types based on the classification by Veron-Cetty & Veron. The y-axis is a log-scale and the cut-off for the Radio-Loud and Radio-Quiet objects lies at $y=1$. As expected, the H2-type galaxies which are dominated by star bursts appear to have systematically lower radio loudness than the S1 and S2 types.

Radio 6cm Luminosities and Radio 20cm Luminosities



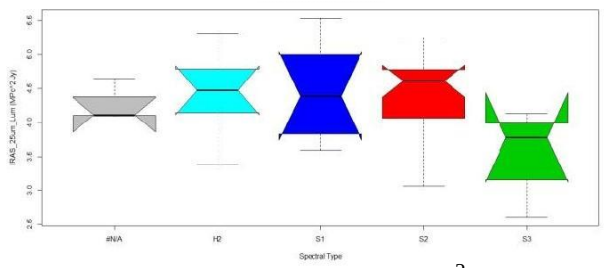
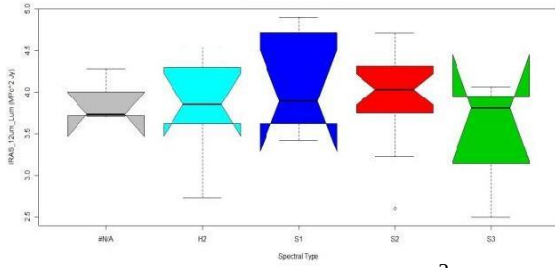
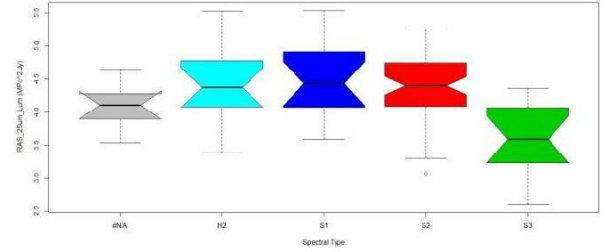
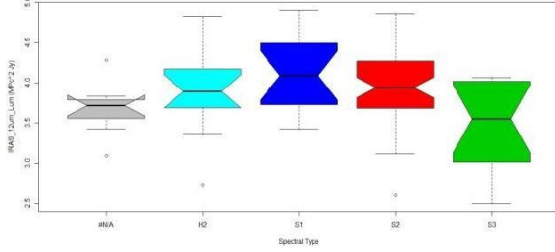
Radio 6cm Luminosity (Mpc² Jy)

Radio 20cm Luminosity (Mpc² Jy)

The Radio Luminosities are plotted for their respective spectral types. The first two box and whisker plots (6cm_All and 20cm_All) are for those objects for which the radio luminosities are available based on the S7 list. There are 84 objects with 6cm luminosity out of which there are 51 radio quiet objects. The lower panel shows the corresponding plots for the radio quiet objects only. Similarly, there are 101 objects with 20cm luminosity values out of which there are 43 radio quiet objects. There seems to be no significant difference between S1 and S2 type objects in both the radio quiet cases. The H2 types in both the radio quiet luminosity plots seem to show marginally lower radio luminosities.

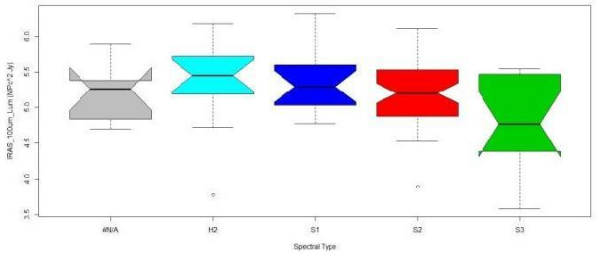
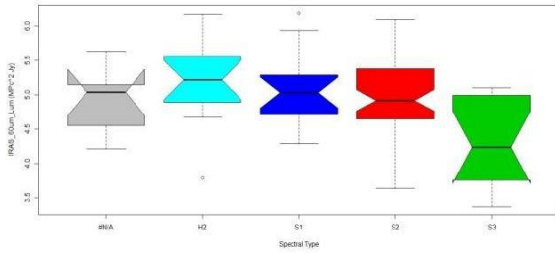
IRAS Luminosities at 12 μ m, 25 μ m, 60 μ m and 100 μ m

Two sets of plots are shown for each infrared luminosity (viz., 12 μ m, 25 μ m, 60 μ m and 100 μ m). In each case the top panel shows the distributions for all the AGNs classified by spectral type, and the bottom panel shows the corresponding distributions for the radio-quiet objects alone.



IRAS 12 μ m Luminosity (Mpc² Jy)

IRAS 25 μ m Luminosity (Mpc² Jy)



IRAS 60 μ m Luminosity (Mpc² Jy)

IRAS 100 μ m Luminosity (Mpc² Jy)

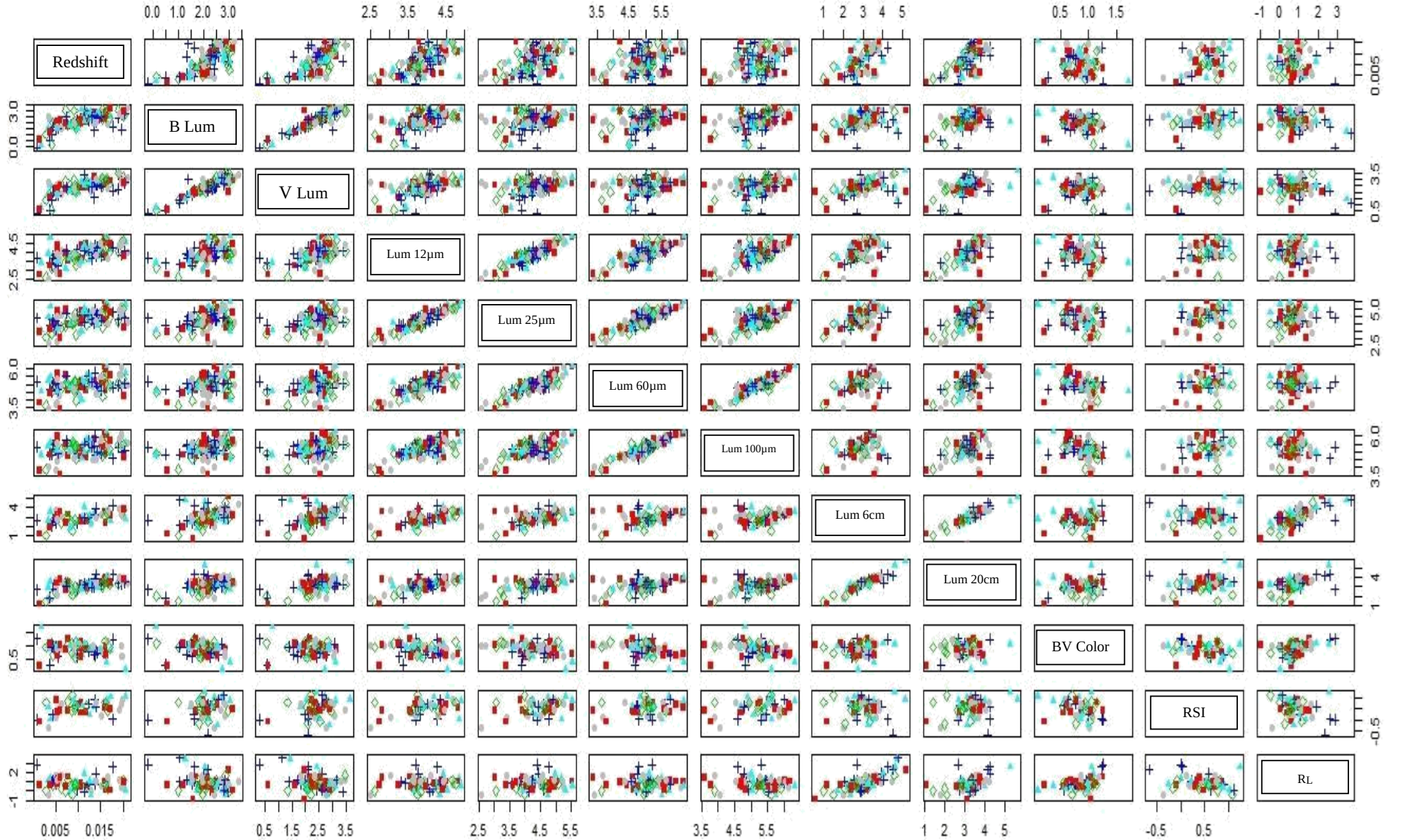
There is a marginal difference between the median values of H2 in the 60 μ m luminosity radio quiet plot as well as in the case H2 of the 100 μ m radio quiet luminosity as compared to S1 and S2 in the respective plots. This implies to higher star formation leading to more cool dust components detected in the infrared.

2. We have plotted two scatterplot matrices (one with Flux densities and the other with the Luminosities with other parameters) for different Spectral-Type with parameters:

- Redshift,
- *B-Flux* density,
- *V-Flux* density,
- IRAS Flux density/Luminosity (12 μ m),
- IRAS Flux density/Luminosity (25 μ m),
- IRAS Flux density/Luminosity (60 μ m),
- IRAS Flux density/Luminosity (100 μ m),
- Radio Flux density/Luminosity (6cm),
- Radio Flux density (20cm),
- *BV Color Index*,
- Radio Spectral Index, and
- Radio Loudness



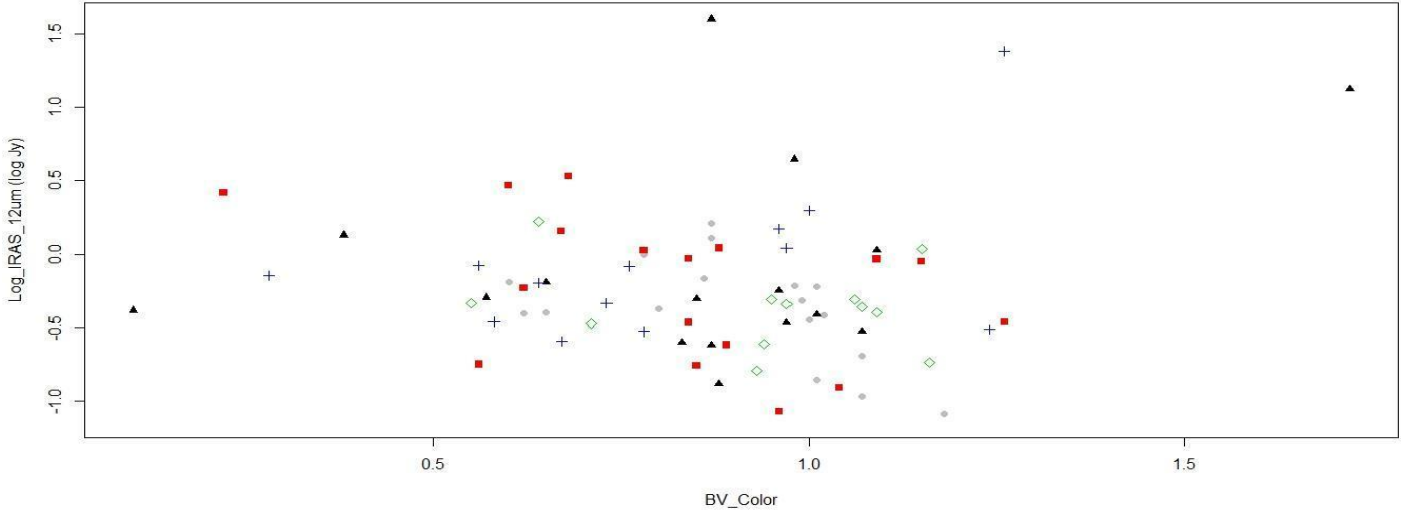
Comparing the **Log_B_Flux** vs **Log_V_Flux** plot, we find that they show a tight correlation as expected with some objects lying outside the trend. For example, there is a “plus-marked” object (a S1 type object) which is on the bluer end of the scale meaning that it might be due to heavy star formation or due to a bright AGN. Similarly there is a “red-square” (a S2 type object) which is towards the redder end of the scale which could be due to dust extinction which may imply that the host galaxy orientation is edge-on. In the IRAS fluxes, suppose we consider the **12 μ m-100 μ m** plot and **60 μ m - 100 μ m** plot. Comparing these two plots we observe that as the difference between the wavelengths increases there is larger scatter in the correlation. In case of the 60 μ m-100 μ m, there seems to be quite a strong correlation for the objects which might be due to the emission mechanisms being similar in those two wavelengths which in the 12 μ m-100 μ m plot shows that for some objects at a given 100 μ m flux density there is an excess of the corresponding 12 μ m flux density. These objects need to be further investigated. There seems to be no significant trend for the **Log_RL** vs the other parameters.



This scatterplot matrix shows similar trends like that in the previous scatterplot matrix. One particular plot of interest here is the **Redshift vs Log_V_Lum**, where we can observe that higher redshifted objects are found to be showing stronger V-band luminosity (it is equivalent to the Absolute Magnitude). This is due to the Malmquist bias. The trend in the Luminosities for Radio, IRAS, etc. seems to be similar to that of the corresponding Flux density plots. Here the **RSI** (Radio Spectral Index) plots show a clumping of the objects around the 0.7 value with is in accord with our assumption while calculating the 6cm radio flux densities for the object which only had the 20cm radio flux densities.

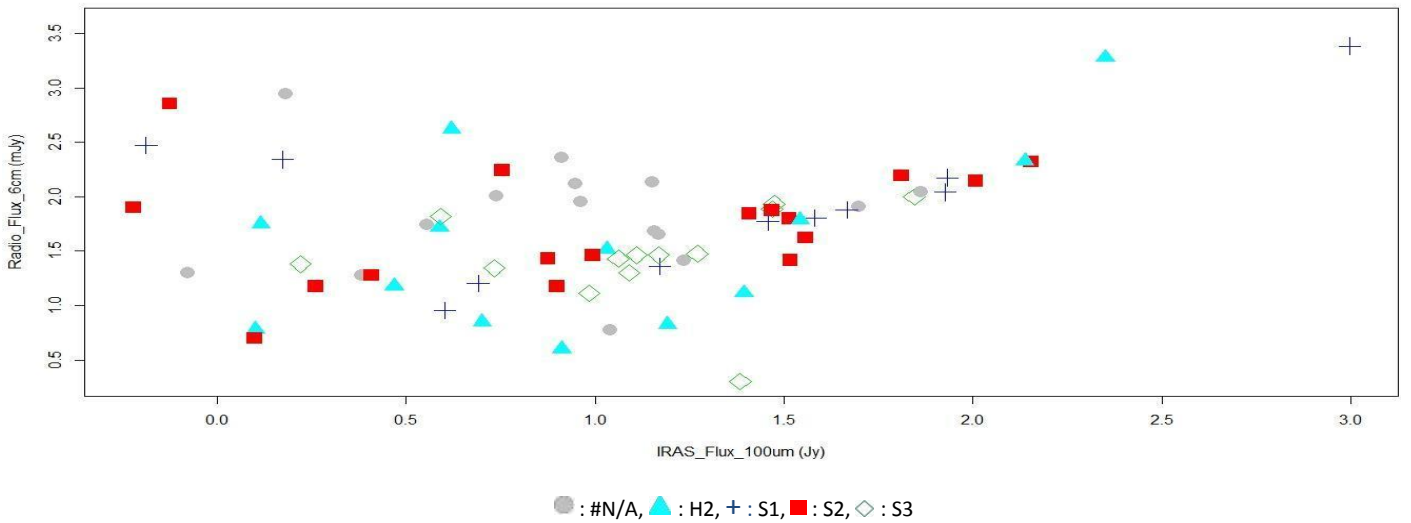
3. Single Scatterplots

IRAS 12 μ m luminosity vs BV_Color:



There is a marginal trend of the bluer galaxies being more luminous in the infrared which may be due to the enhanced star formation giving rise to both blue color and the enhanced infrared emission from the dust. A similar trend is observed for the IRAS 25 μ m, 60 μ m, 100 μ m luminosities vs *BV_Color*.

Radio 6cm flux density vs IRAS 100 μ m flux density:



In this scatterplot, due to the applied cut-off for the Radio flux density ($> 20\text{mJy}$) we initially are observing a flat trend which of course is artificial. The actual trend begins at $x=1.0$ and extends till $x\sim 2.2$ which is due to the radio-infrared correlation. Also for a given value for the IRAS 100 μ m flux density at the lower levels there is radio excess for some the objects which need to be investigated.

CONCLUSIONS

We have attempted to characterize the physical properties for the S7 sample chosen from the Veron-Cetty & Veron Catalogue by compiling the flux density measurements at 6cm and 20cm (Radio) and 12 μ m, 25 μ m, 60 μ m, and 100 μ m (IRAS). Flux density measurements for the B-band and V-band were also taken. We used these measurements to calculate the corresponding values of luminosities, radio spectral indices and radio loudness.

We examine the distributions of the various parameters as well as their pair-wise correlations. From this analysis we find the following

- There is no significant difference in the radio luminosity between the spectral types S1 and S2
- The H2-spectral types appear to have systematically lower radio luminosities as well as radio-loudness, which is not surprising since the emission from the H2-type galaxies are dominated by star-formation
- While there is no significant systematic difference in the infra-red luminosities of the S1 and S2 types, the H2 galaxies show systematically higher 60 μ m and 100 μ m luminosities indicative of large amounts of cool dust.
- The redshift-luminosity correlations clearly show that Malmquist bias affects the sample even though the redshift upper-limit of the S7 sample is extremely low (0.02).
- While the luminosities from nearby infra-red bands (12 μ m-25 μ m and 60 μ m-100 μ m) are correlated, the scatter in the 12 μ m-60 μ m or 12 μ m-100 μ m is much larger, indicative of dusts at different temperatures contributing in the different cases, and possibly AGNs contributing to the 12 μ m excess.
- There is a marginal trend for bluer galaxies to have higher infrared luminosities, consistent with enhanced star formation giving rise to both bluer emission and higher emission from dust.
- Beyond the radio-flux density cut-off, the sample objects are broadly consistent with the well-known radio-infrared correlation for galaxies (*Dwek & Barker 2002*), although several objects show a radio excess, possibly due to the AGN emission.

Future prospects with S7:

- To refine the classification based on morphology of the host galaxy and spectra
- To relate the observed AGN properties to the host galaxy orientation
- To compile the super-massive black holes mass estimates
- To get the accretion rates from the X-ray fluxes for these objects
- This classification would serve as a superset from which smaller samples can be drawn for specific studies like comparing Seyfert Type 1 and Seyfert Type 2 objects with based on similar face-on orientations and at comparable redshifts.

REFERENCES

1. *Antonucci I. (1993)*, Unified models for active galactic nuclei and quasars, Annual review of astronomy and astrophysics. Vol. 31 (A94-12726 02-90), p. 473-521.
2. *Blandford R.D., Netzer H. and Woltjer L. (1990)*, Active Galactic Nuclei (Springer-Verlag).
3. *Dopita M., Rhee J., Farage C. (2010)*, The Wide Field Spectrograph (WiFeS): performance and data reduction, et al., Astrophysics & Space Science, Vol. 327, Issue 2, pp. 245-257.
4. *Doyle, D.* (<https://sites.google.com/site/davidsstatistics/home/notched-box-plots>)
5. *Dwek E., Barker M.K. (2002)*, The Cosmic Radio and Infrared Backgrounds Connection, The Astrophysics Journal, 575:7-11.
6. *Krolik J.H. (1999)*, Active Galactic Nuclei – from the Central Black Hole to the Galactic Environment (Princeton Series in Astrophysics).
7. *Urry C.M., Padovani P. (1995)*, Unified Schemes for Radio-Loud Active Galactic Nuclei, Publications of the Astronomical Society of the Pacific, Vol. 107, p. 803.
8. *Windhorst, R. A.; Fomalont, E. B.; Kellermann, K. I.; Partridge, R. B.; Richards, E.; Franklin, B. E.; Pascarelle, S. M.; Griffiths, R. E. (1995)*, Identification of faint radio sources with optically luminous interacting disk galaxies, Nature, Vol. 375, Issue 6531, pp. 471-474.
9. *Winkler H., Zorroaster* (<http://www.uj.ac.za/EN/Faculties/science/departments/physics/Pages/ZORROASTER.aspx>)
10. *Wright N., Cosmological calculator* (<http://www.astro.ucla.edu/~wright/CosmoCalc.html>)
11. R-Project (<http://www.r-project.org/>)
12. TopCat (<http://www.star.bris.ac.uk/~mbt/topcat/>)